

PAPAYA VL

Datasheet

Electrical characteristics, ratings and hardware features.

V1.0.0



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Chapter 1

About this datasheet

This document gives the electrical characteristics, ratings and hardware features of the **PAPAYA VL** board. It is the reference for selecting the board, wiring it, and bringing it up.

It does not describe the software. For designing and deploying block diagrams see *PAPAYA Studio – User Manual*; for the host APIs see *PAPAYA Libraries – C++ and Python Reference*; for the wire protocol see *DLP Protocol – Programmer Manual*.

Note

Read the absolute maximum ratings before connecting anything. Exceeding any of them can damage the board permanently, and several limits are lower than a first guess would suggest.

Chapter 2

Overview

PAPAYA VL is a mixed-signal control, audio and motion development board. A block diagram designed on a host is deployed to the board, which executes it in hard real time at a configured sampling frequency. There is no firmware to write for the board and no toolchain to install for it.

2.1 Feature summary

Domain	Provision
Analog inputs	10 channels, 16-bit, -10 V to +10 V
Analog outputs	5 channels, 12-bit, -10 V to +10 V, with a waveform generator
Comparator inputs	5 channels, -10 V to +10 V
PWM outputs	8 , independent
PWM inputs	5 channels, two pins each
Motor drive	2 brushed-DC power stages with current measurement
Encoders	2 incremental, A/B, 5 V and 3.3 V
Audio in	2 microphone channels on one I2S interface
Audio out	2 independent mono power stages
Motion sensing	9 axes: acceleration to +/-2 g , rate of rotation to +/-250 deg/s , and magnetic field, 3 axes each
External sensors	4 SPI ports, each with its own chip select. 4 I2C connectors, all on one bus
Non-volatile storage	256 KB for stored designs, in up to 16 slots. A separate 8 MB store holds one audio recording and 8 signal datasets

Domain	Provision
Host link	USB device, or SPI slave
Stacking	up to 4 boards on one SPI master
Form factor	Arduino-header compatible
Power	External DC input. The board is not powered over USB
Design capacity	16 scalar probes, 9 vector probes and 16 remotely-writable inputs, at every sampling frequency

2.2 Applications

Motor control - robotics - condition monitoring - biomedical signal analysis - flight control - audio processing - control-systems teaching.

Chapter 3

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. These are stress ratings only; operation at these limits is not implied.

Parameter	Min	Max	Unit	Note
DC input voltage, PWR	0	16	V	see the note below
Analog input voltage, AIN1-AIN10	-10	+10	V	
Comparator input voltage, CMP1-CMP5	-10	+10	V	
Frequency-input voltage, FI channel 1 pins	0	3.3	V	no series protection
Total continuous DC input current	–	4	A	set by the input protection device
Operating ambient temperature	0	+70	degC	

Note

The FI channel-1 pins connect directly to a processor GPIO with no series resistor, clamp or buffer. They are **3.3 V logic**. Over-driving one damages the board.

Total input current is limited to 4 A, and both motor stages draw through it. This is the binding constraint on what the board as a whole can deliver, whatever any individual stage is rated for.

Chapter 4

Power

4.1 The board is not powered over USB

A DC supply on PWR is required. The USB connection carries the host link and nothing else: its bus voltage reaches only the USB interface, through a current-limited switch, on a node that never joins the board's own 5 V rail.

A board with only USB plugged in enumerates on the host and does not run. This is the first thing to check when a board appears dead, and it is the single most common bring-up mistake.

4.2 Supply and protection

PWR is a two-position pluggable terminal block, positive on pin 1 and ground on pin 2. Between it and the rest of the board sit, in order: a self-resetting fuse, a bidirectional transient suppressor, reverse-polarity protection, an inrush limiter, and the power switch.

What that chain means in practice:

- **A reverse-connected supply does not damage the board.** It does not run either; correct the polarity and it starts.
- **The input fuse resets itself.** An overload opens it and it recovers once the fault is removed and the board has cooled. There is nothing to replace.
- **The 4 A limit is the whole board's,** motors included. It is a protection rating, not a consumption figure.

4.3 Rails available to a design

Rail	Rating	Notes
5 V	2 A	Switching regulator. Present on J2
3.3 V	see the note below	Linear, derived from the 5 V rail. Present on J2
Analog supplies	200 mA each	Enabled by software, not at power-up
Audio, per channel	5 V, 1.5 A	One dedicated supply per speaker channel
Motor supplies	the PWR input voltage	Motors run at the input rail

Note

The 3.3 V rail is linear, so its current is also 5 V current. The difference is dissipated as heat on the board: drawing 1 A at 3.3 V spends about 1.7 W doing nothing useful. The silicon will pass considerably more than that. Treat a few hundred milliamps as a comfortable working budget and take a larger 3.3 V load from your own regulator off the 5 V rail.

The analog supplies are switched on by software, not by the power switch. Probing an analog output before a design is running measures nothing. That is not a fault.

Chapter 5

Electrical characteristics

5.1 Analog inputs

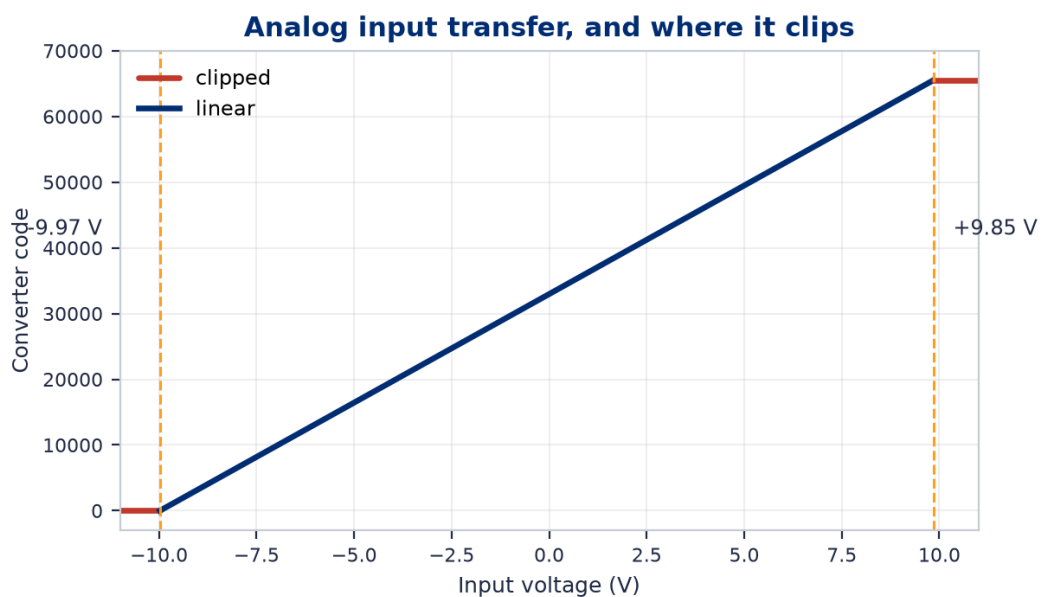


Figure 5.1: Analog input transfer. The converter reaches full scale at approximately -9.98 V and +9.85 V; inputs beyond those points clip rather than wrap.

Parameter	Symbol	Min	Typ	Max	Unit	Conditions
Number of channels	–	–	10	–	–	AIN1 - AIN10
Input range, nominal	V_{IN}	-10	–	+10	V	
Full-scale tolerance	–	–	± 1.5	–	%	of full scale
Resolution	N	–	16	–	bit	over the full span

The block output is in volts. With the default scale of 1.0 and offset of 0.0, an analog-input block reads volts directly; there is no converter code to convert and no bit depth to divide by. The scale and offset exist to express a reading in a *sensor's* units – for a sensor giving 100 degC per volt, set the scale to 100.

Note

Full scale is reached at approximately **-9.98 V** and **+9.85 V**. Inputs beyond those points clip rather than wrap.

5.2 Analog outputs

Parameter	Symbol	Min	Typ	Max	Unit	Conditions
Number of channels	–	–	5	–	–	A01 - A05
Output range	V_{OUT}	-10	–	+10	V	
Resolution	N	–	12	–	bit	
Step size	–	–	4.9	–	mV	
Waveform frequency	–	–	–	10	kHz	generated waveforms

The five outputs are not interchangeable. A01, A04 and A05 share one trigger timer, so they cannot hold different update rates: setting the rate of any one of the three changes the other two, including channels already running. Only A02 and A03 own a private timer.

A04 and A05 also reach their pins through an on-chip buffer amplifier rather than directly, and their transfers are served at a lower internal priority. The voltage range is identical on all five; the settling and noise behaviour is not.

Note

If two outputs must run at different update rates, put one of them on A02 or A03.

What the generator will and will not accept. A generated waveform is played from a stored table of at most 2048 samples, at up to 1 000 000 samples per second, with at least 8 samples per cycle. Those three numbers bound the repetition rate at 125 kHz, and PAPAYA Studio refuses a faster one outright rather than playing something else. They also bound a pseudo-random binary sequence, which has to fit the table whole: at a clock divider of 1 that is 11 bits, and fewer as the divider rises.

The 10 kHz row above is a different figure from that 125 kHz ceiling, and the gap between them is listed under *Open items*.

Every generated waveform repeats. Playback from the table is circular and runs until it is stopped, so a Step becomes a repeating edge, an Impulse becomes an impulse train, and a chirp restarts once per table. If you want a single event, stop the output after it. Stopping parks the channel at 0 V.

Out-of-range waveforms flat-top, they do not wrap. Clamping is applied to each sample as it is played, so a waveform whose peak exceeds the rail comes out clipped at the rail rather than folded back through it.

Generator parameter	Value
Table depth	2048 samples per channel
Table playback rate	up to 1 000 000 samples per second
Minimum samples per cycle	8
Maximum repetition rate	125 kHz, from the three limits above
User-supplied buffer length	exactly 100 samples
Multisine lines	up to 64
Pseudo-random binary sequence	2 to 15 bits, and the whole sequence must fit the table
A DC shape	takes the offset and ignores the amplitude

5.3 Frequency and comparator interfaces

Parameter	Value	Note
Frequency outputs	8, independent	F01 - F08
Frequency inputs	5 channels	FI11/FI12 ... FI51/FI52
Pins per frequency input	2	channel 1 and channel 2 of the same input
Comparator inputs	5	CMP1 - CMP5

5.3.1 Frequency output characteristics

Parameter	Min	Typ	Max	Unit	Conditions
Carrier frequency	0.064	1000	137 500 000	Hz	per channel, set independently
Duty resolution at 1 kHz	–	55 000	–	steps	
Duty resolution at 20 kHz	–	13 750	–	steps	

Parameter	Min	Typ	Max	Unit	Conditions
Duty resolution at 1 MHz	–	275	–	steps	
Maximum duty at 20 kHz	–	99.988	–	%	never exactly 100 %
Output levels	0	–	3.3	V	buffered logic

Duty is a fraction, not a percentage. 0.0 is off, 1.0 is fully on, and 0.25 is a quarter. Nothing anywhere converts between the two conventions, so a value of 50 does not mean half, it means fully on.

Duty resolution falls as the carrier rises, because a period is divided into a whole number of steps. A requested duty is truncated to the step below it rather than rounded. PAPAYA Studio warns when a carrier leaves fewer than 100 steps in a period, which happens above roughly 2.75 MHz.

Realisable carriers are quantised for the same reason. Studio warns when the nearest achievable carrier is more than 0.5 % from the one asked for, and names the frequency the board will actually run.

There is no dead-time insertion and no complementary output pairing. Two channels driving the two halves of a bridge will overlap unless the design prevents it.

The eight frequency outputs are independent. Each carries its own carrier, fixed when the design is deployed, and there is no coupling between them.

A frequency input measures one quantity at a time – pulse width *or* period *or* delay – so pulse width and period cannot be measured simultaneously on one channel. There is no built-in duty-cycle readout for that reason.

A frequency input reports a time in seconds, whichever mode it is in: the high time of a pulse, the length of a period, or the delay between edges on the two pins. It does not report a frequency; take the reciprocal of the period if that is what you want.

Parameter	Value	Note
Measurement range	set per channel, default 1 s	The range, not a gain
Update rate	the input signal's own rate	Not the design's sampling rate
Reading before the first edge	0	
Reading after the signal stops	the last measurement, held indefinitely	No timeout and no error
Interval longer than the range	wraps	The reading is the true time modulo the range
Input glitch filter	on, at its strongest setting	Narrow spikes are ignored

Parameter	Value	Note
Unconnected input	reads low	A weak pull-down holds the pin

Note

A stopped signal looks like a steady one. Nothing times out, so a stalled shaft and a shaft turning at exactly the last measured speed produce the same reading. If that distinction matters, measure it in your design: watch for a reading that has not changed for longer than it should have.

Only the period measurement is available from a Model Builder design today. A pulse-width or edge-delay capture produces two readings on the board, the measurement and the period of the same signal, and the board starts a block only once every reading it produces is connected; the block carries one output port, so the second reading is left unconnected and the block never runs. Such a design uploads, reports success and reads 0. PAPAYA Studio warns about it before it deploys. Period capture is unaffected, and simulation runs all three.

Putting a frequency input into comparator mode claims the analog output of the same channel number as its threshold reference; that output is then no longer free.

Note

The two pins of a frequency input are two *sub-channels of one input*, not two independent inputs. FI52 is Freq In **5**, sub-channel **2**. Channel 1 is the pin every mode measures; channel 2 is used only for edge delay, or driven by the comparator.

5.4 Motor drive

Parameter	Value	Note
Channels	2	MOTOR 1, MOTOR 2
Driver stage rating	6 A continuous, 10 A peak per channel	rating of the driver stage
Board DC input limit	4 A total	input protection; both motors draw through it
Armature current measurement	approximately ± 5.3 A	measurement span
Supply	board input voltage	motors run at the input rail

Note

The three figures above are not alternatives; read them together. The driver stage is rated for 6 A, the board's own input protection admits 4 A in total, and the current measurement

spans about ± 5.3 A. **The board cannot deliver 6 A per channel through its own input**, and a current beyond the measurement span cannot be observed.

Over-current protection is not available. The motor current watchdogs are future work. Do not design a system that relies on the board to limit motor current.

5.4.1 Motor drive characteristics

Parameter	Min	Typ	Max	Unit	Note
Commanded duty	-1.0	–	+1.0	–	signed fraction; the sign is the direction
Drive carrier frequency	0.064	20 000	137 500 000	Hz	
Duty resolution at 20 kHz	–	8500	–	steps	falls as the carrier rises
Current measurement resolution	–	2.67	–	mA	per code
Current measurement span	-5.47	–	+5.47	A	about a calibrated zero
Control and measurement rate	100	1000	1000	Hz	see below
Drive-enable threshold on the motor supply	2.4	–	–	V	

Duty is signed and the sign is the direction. A duty of -0.4 drives the same magnitude as +0.4, the other way.

The current measurement zeroes itself after every start. The board skips the first few control periods and then averages a block of readings to establish zero, which takes roughly the first hundred milliseconds. Hold the duty at zero over that window or the zero is taken against a real current.

Without motor power the drive stays disabled and says nothing. Commands are accepted, the current probes read zero and no error is raised. If a motor does not turn, check the motor supply before the design.

Motor and encoder measurements update at the control rate, which follows the design's sampling frequency but is clamped between 100 Hz and 1000 Hz. A design running faster than 1 kHz reads the same motor and encoder values in consecutive periods.

A MOTOR block and the ENCODER block of the same number cannot both be used, because the motor stage owns that encoder. A motor block configured for current measurement only releases it.

5.5 Encoders

Parameter	Value
Channels	2 (ENC 1, ENC 2)
Signals	A and B only
Logic levels supported	5 V and 3.3 V
Decoding	quadrature, 4 counts per line

Note

There is no Z / index line on either encoder connector. Index-based homing is not possible. Establish a zero with a limit switch, a mechanical stop, or by convention at power-up.

Position is a free-running count with no absolute zero; it starts from whatever the counter held.

Parameter	Value	Note
Counts per line	4	both edges of both channels
Position counter	32-bit signed, -2 147 483 648 to +2 147 483 647	
Measurement update rate	100 Hz to 1000 Hz	the motor and encoder control rate
Input pull-ups	10 kohm to the selected encoder supply	open-collector encoders work directly
Logic level selection	one switch for both connectors	5 V or 3.3 V; the two cannot be mixed

A 1000-line encoder therefore gives 4000 counts per revolution.

Position is zeroed by a fresh deployment, and by nothing else. Stopping and restarting sampling within one deployment leaves the count where it was.

Velocity is derived from the interval between edges and smoothed by a first-order filter whose time constant is a block parameter. Two consequences follow. When the edges stop, the reading decays toward zero rather than snapping to it; on the motor path it is forced to exactly zero after about 385 ms without an edge. And the motor path declares a stall when the position has moved

by two counts or fewer over about ten milliseconds, and forces the velocity to zero. A standalone encoder block has no stall detector.

Note

The two encoder connectors share one voltage-select switch. A 5 V encoder and a 3.3 V encoder cannot be used at the same time.

5.6 Audio

Parameter	Value	Note
Microphone channels	2	MIC-L, MIC-R connectors, one shared I2S bus
Microphone interfaces	1	both channels share one I2S bus
Speaker channels	2, independent mono	not a stereo pair
Speaker current	1.5 A continuous per channel	
Audio sample rate	44 100 Hz, fixed	not derived from the design rate

Note

The audio interfaces are clocked at a fixed 44 100 Hz regardless of the design's sampling frequency, and nothing resamples. Run audio designs at 44 100 Hz; at any other rate the design and the audio buffer run at different speeds.

The two speaker outputs are bridge-tied. Never connect them together, and never connect either output to ground.

Parameter	Value	Note
Sample format	16-bit	both directions
Signal range in a design	-1.0 to +1.0	normalised, not volts
Speaker output buffer	128 frames	about 2.9 ms, on top of the one-period actuation delay
Out-of-range speaker input	hard-clipped	
Microphone channels per period	both	the pair is read together, sample-simultaneous

A speaker block is mono and is played on both halves of the frame. The two speaker channels are two independent mono outputs, not a stereo pair.

The microphone stream can carry a DC offset. Remove it in the design if it matters; nothing removes it for you.

Note

The microphone connectors supply power to the microphone module, and the board also drives the left/right select line. Do not drive that line back into the board.

5.7 Inertial sensing

Sensor	Axes	Output data rate	Anti-alias filter
Accelerometer	3	200 Hz	50 Hz
Gyroscope	3	400 Hz	100 Hz
Magnetometer	3	100 Hz	–

Sensor	Full scale	Beyond it
Accelerometer	+/-2 g , fixed	clips
Gyroscope	+/-250 deg/s , fixed	clips
Magnetometer	0.15 microtesla resolution	–

The full-scale ranges are fixed and cannot be changed. A design that has to survive a knock beyond 2 g, or a rotation faster than 250 degrees per second, will read a clipped value rather than a large one.

All nine axes share one frame, the triad printed on the board: X across the board, Y up it, Z out of the face. Nothing is wired to any of them, so orientation is the only thing a design has to get right.

Note

The output data rates above are the real bandwidth ceiling, and they are fixed. Raising the design sampling frequency past them does not buy more motion bandwidth – it re-reads the same registers.

Readings reach a design already in engineering units: acceleration in g, angular rate in degrees per second, magnetic field in microtesla.

Nothing is corrected on the board. There is no bias removal, no temperature compensation and no cross-axis correction. A stationary gyroscope reads a small non-zero rate, and that bias belongs to your design to estimate and subtract.

5.8 External sensor ports

Parameter	Value
SPI ports	4, each with its own chip select
I2C connectors	4, all on one bus; give each device a different address
I2C bus pull-ups	on the board, one 10 kohm to 3.3 V on each of SDA and SCL

A sensor that fails to start is refused at deployment, with a message naming it, rather than running a design that reads nothing.

A sensor that fails while running holds its last value through a transient glitch on the bus. After ten consecutive failed reads the board drops the device and its outputs read zero until the design is deployed again.

The time a bus transaction takes bounds the sampling frequency a design with external sensors can reach.

Chapter 6

Non-volatile storage

The board carries two separate non-volatile stores. Both keep their contents with no power applied.

6.1 Stored designs

Parameter	Value
Total capacity	256 KB (262 144 bytes)
Reserved for the directory	4096 bytes
Available to designs	258 048 bytes, shared across all slots
Number of slots	16
Label length	up to 128 characters

A stored design lets the board run standalone at power-up, which is what turns a PAPAYA VL from a tethered demonstrator into a deployed instrument. One of the sixteen can be marked to load automatically on reset.

The 258 048 bytes are a shared pool, not a per-slot allowance: sixteen small designs fit comfortably, and one very large design can occupy most of the space on its own.

6.2 Recordings and datasets

Parameter	Value
Total capacity	8 MB
Audio recording slots	1
Audio capacity	4 128 768 bytes, 1 032 192 samples

Parameter	Value
Audio duration	about 23 s at 44 100 Hz, about 129 s at 8000 Hz
Dataset slots	8
Dataset capacity, each	458 752 bytes, 114 688 samples
Sample format, both	32-bit floating point

A dataset is a recorded signal a design can play back as a source, so a test that needs a real-world input does not need the real world attached. The audio slot holds one recording, which a design can play or a host can read back.

Note

A live recording is bounded by a much smaller staging area than the slot it ends up in. A recording made by a running design is limited to 65 536 samples, which is about 1.5 s at 44 100 Hz and about 8 s at 8000 Hz. When that fills, the recording keeps its beginning and discards the rest. The much larger figures above apply to what the slot can hold, which is reached by uploading a recording from the host rather than by capturing one live.

Chapter 7

Mechanical

All dimensions are in millimetres, measured from the bottom-left corner of the board outline looking at the component side.

7.1 Outline

Board outline	157.0 x 90.0
Corner radius	1.27 (all four corners)
Left-edge recess	12.0 deep x 34.0 tall, spanning y = 28.0 to y = 62.0
Plan envelope including connector overhang	165.4 x 94.8
Height envelope	23.95 overall, from 13.95 above the board to 10.00 below it

The recess in the left edge is a cut-out in the outline itself, not a keep-out: nothing is fitted there. The plan envelope is larger than the outline because four connectors overhang the edge, the largest by 4.94 mm; allow for the envelope rather than the outline when the board goes into a housing. The height envelope is set by the stacking pins, which stand 13.95 mm proud on top and 10.00 mm below.

7.2 Mounting holes

Four plated holes, one near each corner.

Hole	x	y
Bottom left	5.2	5.2

Hole	x	y
Top left	5.2	84.8
Bottom right	145.5	5.2
Top right	145.5	84.8

The four centres span 140.3 x 79.6. Each sits 5.2 from the left, bottom and top edges; the right-hand pair sits 11.5 from the right edge, not 5.2, so the pattern is not symmetric about the board centre. Each hole has a 6.0 x 6.0 pad and takes an M3 screw with clearance.

Chapter 8

Real-time execution and design capacity

8.1 Sampling frequency

The design's sampling frequency is a single global setting. Every block runs once per period, in an order the board determines from the wiring.

Parameter	Value
Accepted range	0.001 Hz to 1 000 000 Hz
Default	1000 Hz
Practical real-time range	about 0.1 Hz to 50 kHz

The board accepts rates it cannot meet in real time. Rather than refusing a frequency outright, it reports the CPU load it actually measures per sampling period, so an over-committed design says so with a measurement instead of a guess. Check that figure before trusting a fast design.

Three parts of the board have their own rate, independent of the design's:

- **Audio runs at 44 100 Hz and nothing resamples**, so an audio design should run at 44 100 Hz.
- **Motor and encoder measurements update between 100 Hz and 1000 Hz.**
- **The inertial sensors deliver new data at the output data rates given above.** Sampling faster re-reads the same values.

8.2 Capacity

Resource	Limit	Note
Scalar probes	16	at every sampling frequency

Resource	Limit	Note
Vector probes	9	
Vector probe frame	8192 bytes, 2048 values	a 2048-point spectrum sits exactly at the ceiling
Remotely-writable inputs	16	counted through macros
Frequency outputs	8	one block each
Frequency inputs	5	one mode at a time each

Note

The probe budget does not change with sampling frequency. Sixteen scalar probes are available at 100 Hz and at 50 kHz alike. What changes with frequency is how much of the data the host link can carry, which the Monitor reports while it runs.

Chapter 9

Connectors and pin assignment

Connector designators are as printed on the board.

Designator	Function
PWR	DC input
USB-C PD	USB host link
J2	Main I/O header – analog, frequency, comparator
J7	SPI host port, in an Arduino shield footprint
MOTOR 1, MOTOR 2	Motor outputs
ENC 1, ENC 2	Encoder inputs
SPEAKER 1, SPEAKER 2	Audio power outputs
MIC-L, MIC-R	Microphone inputs, both on one I2S bus. The L-R way is held by the board, 3.3 V on MIC-L and 0 V on MIC-R, so the two take opposite halves of the frame
SPI 1 - SPI 4	External SPI sensor ports
I2C 1 - I2C 4	External I2C sensor connectors. Four connectors, one shared bus : give each device a different address. The board carries the bus pull-ups, one 10 kohm to 3.3 V on SDA and one on SCL, shared by all four connectors
SPI SLAVE ADDRESS SEL	Board address 0-3 when stacked
GAIN SEL 1, GAIN SEL 2	Speaker gain selection
RESET	Reset

9.1 J2 pin assignment

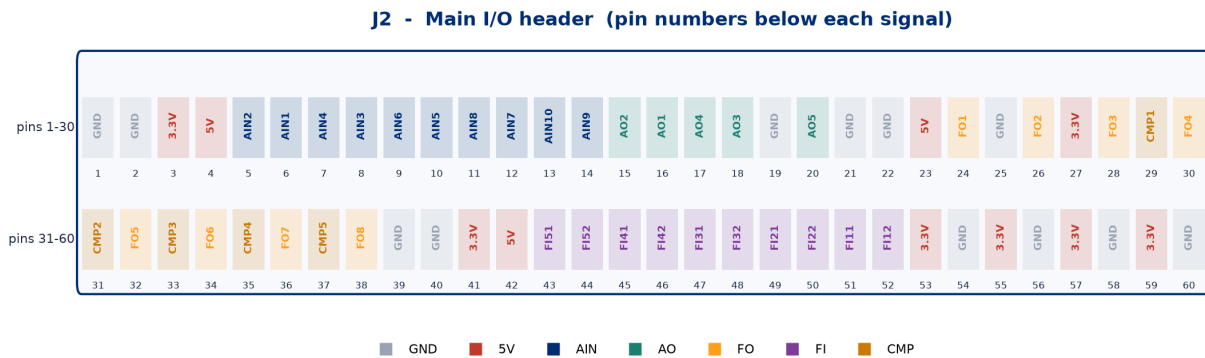


Figure 9.1: J2, the main I/O header. Signal names are the silkscreen designators; the number below each one is the pin number. Note that channel numbering runs against pin parity - pin 5 is AIN2 and pin 6 is AIN1.

9.2 Signal naming

The silkscreen designators are the names to use; they are what is printed beside each pin.

Silkscreen	Meaning
AIN1 - AIN10	Analog input 1 to 10
A01 - A05	Analog output 1 to 5
F01 - F08	Frequency output 1 to 8
CMP1 - CMP5	Comparator input 1 to 5

Each frequency input occupies two adjacent pins on J2, printed FI11 and FI12 for channel 1 and so on to FI51 and FI52. Read the code as FI plus channel number plus sub-channel: FI52 is frequency input **5**, sub-channel **2**.

Studio addresses them as FI1_PULSE, FI1_PERIOD and FI1_DELAY through FI5_DELAY: one physical channel, measured three ways, one way at a time. The printed designator names the pin, the identifier names the measurement.

Note

Channel 1 of every frequency input is unbuffered 3.3 V logic. Do not exceed 3.3 V and do not drive it negative. An overvoltage on these pins damages the board rather than merely misreading.

J7 is an SPI host port in an Arduino shield footprint. A master drives the board over D8 to D13 and resets it over D2. **A0 to A5 are not connected, and neither is any other pin the legend prints.** A shield that uses them will seat perfectly and do nothing.

Chapter 10

Host interface

Parameter	Value
USB	device mode, USB 2.0 High Speed
USB connector	reversible; either orientation works
SPI	slave mode, mode 0, MSB first
SPI clock, default	8 MHz
SPI clock, tested ceiling	40 MHz
Chip-select guard	at least 100 microseconds; 200 microseconds recommended
Boards per SPI master	up to 4, addressed 0 to 3
Reset pulse	100 ms asserted
Time from reset to ready	allow 3 s

Note

The USB-C PD legend does not mean USB Power Delivery. The connector is a USB-C receptacle wired as a fixed sink. There is no power negotiation and no alternate mode, and the board does not draw its operating power from it in any case. See *Power*.

10.1 Driving the board over SPI

The SPI port is a slave. Your master supplies the clock and the chip select, and the board answers.

Allow a guard period around every chip-select edge. The board needs time between the chip select going active and the first clock edge, and again after the last one. Below about 100 microseconds command headers are corrupted; 200 microseconds is the value the supplied host libraries use and is the figure to start from. This is a real arming requirement, not a courtesy delay.

Address each board on the stack with the SPI SLAVE ADDRESS SEL switch, 0 to 3. Four boards can share one master. An address that does not match the one the host is talking to is the failure that nothing else diagnoses: the board is present, powered and silent.

10.2 Monitoring and streaming

Probe data reaches the host in blocks rather than one sample at a time.

Parameter	Value
Block size over USB	128 samples per probe, fixed
Block size over SPI	negotiated, 1 to 128 samples per probe
Sample format	32-bit floating point

The first block of a scalar probe arrives after 128 samples, so at the design's own sampling frequency the wait is 128 divided by that frequency. At 1000 Hz that is 128 ms and invisible. At 2 Hz it is just over a minute of an apparently empty display. That is the block size, not a fault.

Chapter 11

Status indicators

Five lamps, and the colour is the reading. Three sit together and report the board; the other two sit beside the motor terminal blocks and report the drive.

Lamp	Colour	What it means
Power	Blue	Input power is present and the power switch is off
Power	Purple	Input power is present and the power switch is on
Ready	Green	The board has power and a program is running on it
Fault	Orange	The board has hit an internal failure
MOTOR 1 direction	Orange / Blue	Motor 1 is turning; the colour is which way
MOTOR 2 direction	Orange / Blue	Motor 2 is turning; the colour is which way

Fault on means the design running on the board is not doing what it was asked to. Each direction lamp is next to the terminals it belongs to, so the lamp and the wiring are read together.

Note

A dark Power lamp with USB plugged in is the expected result, not a fault: the lamp follows the DC input, which is what powers the board. See *Power*.

Chapter 12

Bringing the board up

In order, for a board out of its box.

1. **Connect a DC supply to PWR**, positive to pin 1. The board does not run on USB alone. Leave the power switch off while you wire it.
2. **Check the polarity before switching on.** Reverse polarity does not damage the board, but it does not run either, and a board that will not start is harder to diagnose than one you have already checked.
3. **Switch on.** The Power lamp turns purple. If it stays blue the switch is off; if it is dark the input is not reaching the board.
4. **Connect the host.** Over USB, plug the cable in either way round. Over SPI, set SPI SLAVE ADDRESS SEL to the address your master will use, and allow the chip-select guard given under *Host interface*.
5. **Deploy a design.** The Ready lamp turns green once a program is running.

Points worth knowing before you wire anything to the header:

- **Channel 1 of every frequency input is unbuffered 3.3 V logic.** Nothing protects those pins. Everything else on J2 is protected.
- **The analog supplies are off until a design enables them.** An analog output measured before deployment reads nothing.
- **J7 accepts an Arduino shield mechanically and connects almost none of it.** See *Signal naming*.

Note

The motor and speaker terminal blocks are the same type and pitch, so a plug from one fits the other. The motor terminals carry the full input voltage. Connecting a speaker to a motor terminal will destroy it. Label your plugs.

Chapter 13

Open items

These figures are not yet established and are deliberately absent rather than estimated:

- **Minimum and recommended DC input voltage.** Only the 16 V maximum is characterised.
- **Audio gain values** for the GAIN SEL positions.
- **USB VBUS load-switch current limit.**
- **Accuracy budget of the armature current measurement.** The transfer is known; the total error budget is estimated at 1.5-3 % and not yet verified.
- **Board thickness.** The sources in hand disagree, so no figure is given: one puts the laminate at about 1.55 mm and another implies 2.18 mm between the silkscreen planes. Settling it needs the stackup or a caliper.
- **Exact mounting-hole drill diameter.** About 3.9 mm, taken from the board rendering rather than from a drill table, which is why it is here and why the mechanical section gives the pad size and the M3 clearance instead.
- **Board weight.** Not recorded anywhere; it needs scales.
- **The Power lamp with USB connected and no DC input.** The lamp is documented above as following the DC input. An earlier revision of this document described a third, red, state for a board running from USB alone; that contradicts the power architecture and is withdrawn pending confirmation on a bench.
- **Maximum encoder count rate.** Not established. The counter is 32-bit and decodes four counts per line; the rate at which it stops keeping up has not been measured.
- **Settling time and noise of the analog outputs,** and the difference between the directly-driven and buffered channels. The difference is real and is described qualitatively above; no figures exist.
- **Microphone sensitivity and the audio input's full-scale acoustic level.**
- **Sustained USB throughput.** The block geometry is given under *Host interface*; the sustained rate has not been measured on the shipped configuration.
- **A usable published current budget for the 3.3 V rail.** The derating that makes the silicon rating misleading is explained under *Power*; the figure that should replace it has not been set.
- **Quiescent, typical and maximum supply current, and power dissipation.** None of the three has been measured, so none is stated. The 4 A input protection limit is a protection rating and not a consumption figure.

- **Storage temperature range, humidity range and a board-level ESD rating.** The operating range of 0 to +70 degC is the only environmental figure characterised. The board carries ESD protection on the USB and input lines, but no board-level immunity level has been established by test, and the frequency-input channel-1 pins are unprotected and would have to be excluded from any such statement.
- **The input voltage ceiling.** The absolute maximum table gives 16 V while the input transient suppressor stands off 13 V. The two are not the same claim and the difference has not been resolved, so design to the lower figure until it is.
- **The generated-waveform frequency figure.** The *Analog outputs* table gives 10 kHz; the generator's own hard ceiling, which PAPAYA Studio enforces, is 125 kHz. Whether 10 kHz is the point beyond which the waveform is no longer characterised, or a figure that has simply not been revisited, is not yet established, so both numbers are stated rather than one of them dropped.